



UNIVERSITY OF SINDH
JAMSHORO

MINI ASSIGNMENT

DECONSTRUCTING HCI & CG

IN DAILY SOFTWARE



DEPARTMENT OF
COMPUTER SCIENCE



NAME

Asif Ali



ROLL NO

2k24/CSE/39



DEPARTMENT

Computer Science



SUBMITTED TO

Sir Rajesh Kumar



DATE

09 - Sep - 2026

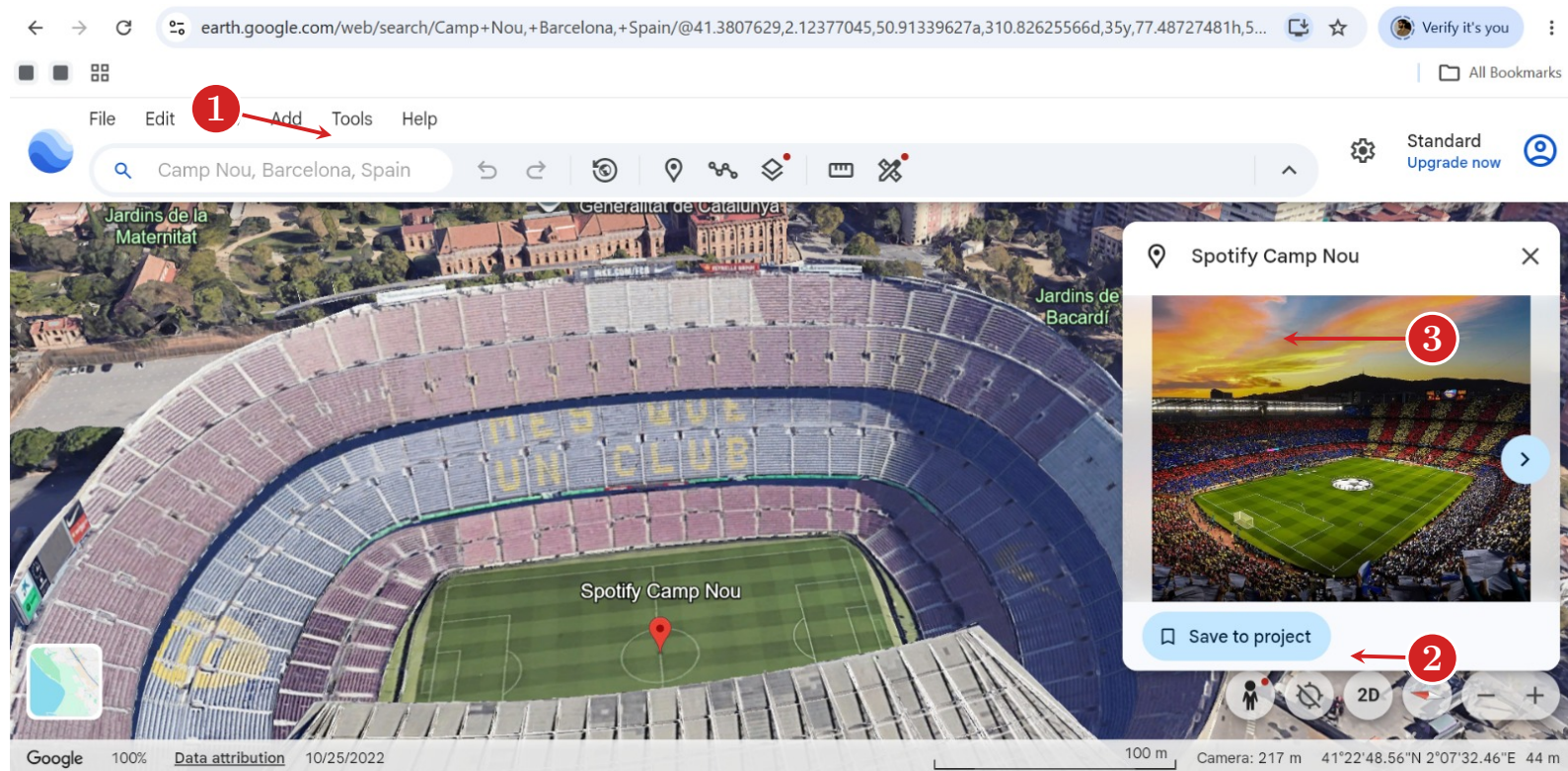
*Technology
Shapes
Better
Tomorrow*

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CODE

💡
CREATE

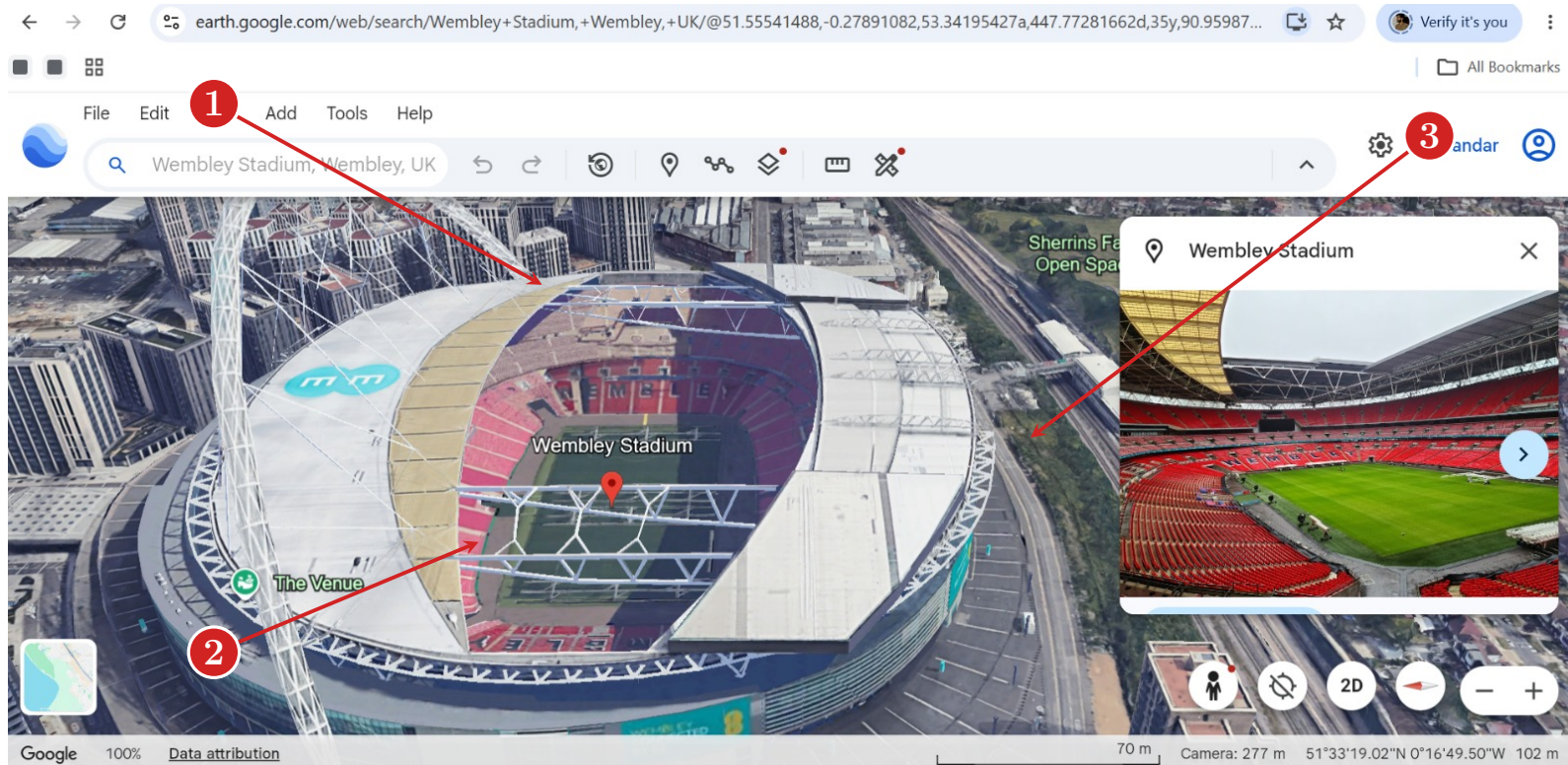
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GROW

SCREENSHOT 1: HCI FOCUS



- 1. Search Bar (Top-Left):** Enables users to search for locations from anywhere in the world. Its autocomplete suggestions appear while typing, which reduces the amount of input required and helps users reach destinations more efficiently.
- 2. 3D Navigation Controls (Bottom-Right):** Provides zoom, rotation, and compass controls for exploring the three-dimensional environment. These controls give users clear visual options for changing and controlling the current view.
- 3. Information Panel (Right-Side):** Shows useful details about the currently selected location, such as its name, description, coordinates, and saving options. This gives users immediate feedback and relevant contextual information.

SCREENSHOT 2: CG FOCUS



- 1. 3D Mesh Geometry:** The Burj Khalifa is represented using a detailed three-dimensional mesh with accurate geometric structure. This allows the building to appear as a realistic 3D object within the virtual environment.
- 2. Texture Mapping:** Detailed satellite imagery is applied to the ground, lake, and nearby surroundings. The texture mapping adds visual detail and creates a more realistic representation of the Dubai landscape.
- 3. Lighting & Shadows:** Dynamic illumination represents the position of the sun and produces shadows across the building and nearby environment. These effects improve depth perception and make the overall three-dimensional scene more realistic.

REFLECTION QUESTIONS

Question 1:

How does the Computer Graphics quality (e.g., realistic lighting or 3D view) help or hinder the user's ability to interact with the software?

The quality of Computer Graphics in Google Earth, such as realistic lighting, detailed 3D models, and texture mapping, improves the user's interaction by making locations easier to recognize and understand. Realistic visual elements help users identify landmarks, judge spatial relationships, and maintain their orientation while exploring the 3D environment. On the other hand, detailed rendering requires greater processing resources. On older or less powerful devices, this may cause reduced performance or lag, which can make interaction less smooth.

Question 2:

What is one simple HCI change you would make to improve the user interface?

A useful HCI improvement would be to introduce a customizable "Quick Actions" toolbar where users could pin the tools they use most often. Features such as measuring distance, opening street view, and enabling 3D buildings can require users to navigate through menus. Placing frequently used functions in a customizable toolbar would reduce the number of clicks and make common tasks faster and more convenient.

Component	Requirement	Points
HCI Annotation	Correctly identifies 3 interaction/UI elements and explains their usability purpose.	35
CG Annotation	Correctly identifies 3 graphics/rendering elements such as lighting, mesh, camera, or related features.	35
Reflection	Provides clear and concise answers connecting graphics rendering with the overall user experience.	20
Clarity & Layout	Presents a clean screenshot layout with readable annotations, callouts, and pointers.	10